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Autumn Term

EC466: Econometrics

Problem Set # 7: Staggered Difference-in-Differences

Background: Consider a policy intervention that was rolled out across different regions at different times. You have panel data from four regions (A, B, C, D) over six time periods (2018-2023), with the following average outcomes:

Region	2018	2019	2020	2021	2022	2023	Treatment Start
A	45.2	46.1	52.8	55.3	57.9	60.2	2020
B	43.8	44.5	45.2	51.7	54.1	56.8	2021
C	44.6	45.3	46.1	46.8	53.4	55.9	2022
D	42.9	43.6	44.2	44.9	45.5	46.1	Never

Each region has 100 observations per period, and treatment is irreversible once adopted.

1. **Two-Way Fixed Effects Estimation**

   (a) Write down the standard TWFE regression specification for this staggered adoption design. Define all variables clearly, including the treatment indicator $\mathcal{D}_{i,t}$.

   (b) Explain intuitively why a researcher might be tempted to use this TWFE specification. What causal parameter might they hope to estimate?

   (c) Calculate three 2×2 DiD estimates "by hand": (i) Region A vs. Region D comparing 2019 to 2020; (ii) Region B vs. Region D comparing 2020 to 2021; (iii) Region B vs. Region A comparing 2020 to 2021. Interpret each estimate.

2. **The Goodman-Bacon Decomposition**

   (a) According to the Goodman-Bacon (2021) decomposition, the TWFE estimator can be written as:

$$\hat{\beta}^{DD} = \sum_{k \neq U} s_{k,U} \hat{\beta}^{DD}_{k,U} + \sum_{k \neq U} \sum_{\ell > k} \left[s^{k}_{k\ell} \hat{\beta}^{DD,k}_{k\ell} + s^{\ell}_{k\ell} \hat{\beta}^{DD,\ell}_{k\ell} \right]$$

Explain what each component of this decomposition represents. What do the subscripts k, U, and ℓ denote?

(b) Describe the three types of 2×2 comparisons that contribute to the TWFE estimator in a staggered design. Which of these comparisons is most likely to be problematic, and why?

(c) The weights in the Goodman-Bacon decomposition depend on sample size and variance in the treatment variable. Explain why the weight $s_{k,U}$ for comparing early-treated group k to the never-treated group U includes the term $(n_k + n_U)^2$. Is this weighting scheme necessarily aligned with policy relevance?

3. **Event Study Specifications**

(a) Write down a TWFE event study regression specification that includes relative time indicators (leads and lags). Explain why a researcher would use this specification instead of a simple TWFE model with a single treatment indicator.

(b) Suppose you estimate a TWFE event study and find that the coefficients on the lead terms (pre-treatment periods) are statistically significant and economically large. Provide two distinct explanations for this finding: one that suggests a violation of identifying assumptions, and one that arises even when all assumptions hold.

(c) Explain why "binning" of endpoints (e.g., using $\mathcal{D}_{i,t}^{<-K}$ for all periods more than K periods before treatment) can contaminate the coefficients on other event-time indicators. What role does treatment effect heterogeneity play in this contamination?